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Website Optimization Experiment Report

Public website versions

V0 baseline: https://liuxinyue84-sketch.github.io/

V1 Few-shot: https://liuxinyue84-sketch.github.io/V1/

V2 RAG: https://liuxinyue84-sketch.github.io/V2/

Executive summary

This experiment evaluates two distinct improvements to Aria's Sky Journal. V0 is the frozen baseline. V1 applies In-context Few-Shot Learning to redesign page and component structure while preserving V0's personal content. V2 independently applies a browser-side Retrieval-Augmented Generation (RAG) workflow to the course-note question module. Both versions were derived separately from V0.

The latest independent AI Judge totals are 81/100 for V0, 85/100 as reported in the V1 raw output (86/100 when its criterion scores are recomputed), and 91/100 for the V2 rerun. V2's public desktop evidence includes 9/9 RAG cases and 13 non-RAG functional checks; the latter recorded one favicon 404. These results support the tested tasks, not arbitrary-query correctness or full mobile accessibility.

1. Task 1: Initial website and V0

The initial prompt asked an AI coding tool to create an English, desktop-friendly personal portfolio with a calm watercolor-sky visual identity, clear sections, basic transitions, and the user-provided biography. Later prompts specified the academic highlights, project contributions, music control for Schumann's Träumerei, a photo and video Life section, and a Guestbook. The preserved prompt appendix distinguishes the original prompt text from reconstructed summaries of unavailable Trae history.

V0 contains Home, Highlights, Projects, Life, and Guestbook. Visitors can browse education and academic records, filter and expand project details, save projects locally, preview photos and videos, control music, search selected course notes, and submit a guestbook entry. Favorites and guestbook entries use browser local storage. Its course-note search is client-side BM25; V0 does not call a live language model.

The baseline was frozen before Task 3. Its full source and static assets are archived under website/V0, with a SHA-256 manifest. The original V0 build prompt and available code excerpts are in prompt_appendix/task1_v0. Some historical Trae prompts and raw outputs were not exported, so reconstructed records are explicitly labelled and are not represented as verbatim logs.

2. Task 2: Independent evaluator

The same version-agnostic rubric was fixed for V0, V1, and V2. Scores use integer anchors from 0 to 5, with 1, 2, and 4 available between the defined levels. The weighted total is $\Sigma(\text{score} \div 5 \times \text{weight})$. The four criteria cover the portfolio's content, visitor tasks, visual readability, and interactive reliability.

Criterion and weight	Definition and reason	0 / 3 / 5 anchors
Content Completeness & Clear Contribution 30%	Checks profile, education, skills, projects, personal/contact content, and whether the owner's contribution is clear. A portfolio must explain who the owner is and what they contributed.	0: most content absent or unusable. 3: key content exists but important gaps or vague contributions remain. 5: coherent required content and clear project contributions.
Navigation & Task Completion 25%	Checks whether visitors can find sections and complete representative tasks such as opening projects, filtering/saving, using the course-note interface, and signing the Guestbook.	0: navigation or most core tasks fail. 3: sections are reachable but important tasks are confusing or incomplete. 5: clear controls support completion of representative tasks.
Readability & Visual Consistency 25%	Checks legibility, hierarchy, spacing, responsive presentation, consistency, and contrast. Design should support comprehension.	0: unreadable or broken layout. 3: recognizable design with material readability or layout problems. 5: clear hierarchy, consistent styling, readable tested viewports.
Interaction Robustness 20%	Checks correct control behavior, state persistence where promised, empty/error handling, and browser/environment constraints.	0: core interactions fail or provide no useful response. 3: common paths work but meaningful edge cases fail. 5: core paths work repeatedly with clear state and useful feedback.

V0 human self-assessment

My fixed-rubric V0 self-assessment is 76/100: Content 4/5, Navigation 4/5, Readability 4/5, and Robustness 3/5. Each example below records the test or inspection, the observed behavior, and why it supports the score.

1. Content — 4/5. Test: I inspected the V0 section structure and four project records. Observation: the site contains the expected portfolio sections, and each project record has purpose and contribution fields. Rationale: the content is substantial, but I had not independently checked every personal claim against the CV/PS or tested whether all details were visible.

2. Navigation — 4/5. Test: I traced the representative path Home → Projects → Research filter → project detail → save. Observation: the navigation anchors, filter, detail control, and local-save handlers are present in the source; the original public desktop run confirmed the main navigation and several interactions. Rationale: usable task paths exist, but not every path, including a valid Guestbook submission, was verified in that assessment.

3. Readability — 4/5. Test: I reviewed the supplied desktop screenshot and shared stylesheet. Observation: the low-saturation sky palette, pill navigation, and content panel form a coherent hierarchy. Rationale: one desktop view supports visual consistency, but does not establish contrast or readability across all sections and viewports.

4. Robustness — 3/5. Test: I inspected local knowledge loading, favorites, Guestbook validation, media, and audio behavior, and considered the dated browser observations. Observation: embedded knowledge

data supports file-based opening, state is stored locally, and the browser may block music autoplay; a MapReduce query also produced a weak leading passage. Rationale: core behavior exists, but relevance and several edge cases remained insufficiently tested.

Independent AI Judge and score comparison

Each version was assessed in a separate blank Judge session using the fixed rubric and version-specific website evidence, without human scores. The original Judge outputs are preserved verbatim. The V1 human score is 86/100 from criterion scores 5, 4, 4, and 4. The V1 AI Judge raw total is preserved as 85/100; its listed criterion scores recompute to 86/100 under the fixed weights. Both raw and recomputed totals are shown in the comparison table.

Version	Human fixed rubric	AI Judge raw total	AI criterion scores	Recomputed AI total
V0	76	81	5, 4, 3, 4	81
V1 Few-shot	86	85	5, 4, 4, 4	86
V2 RAG	86	91	5, 5, 4, 4	91

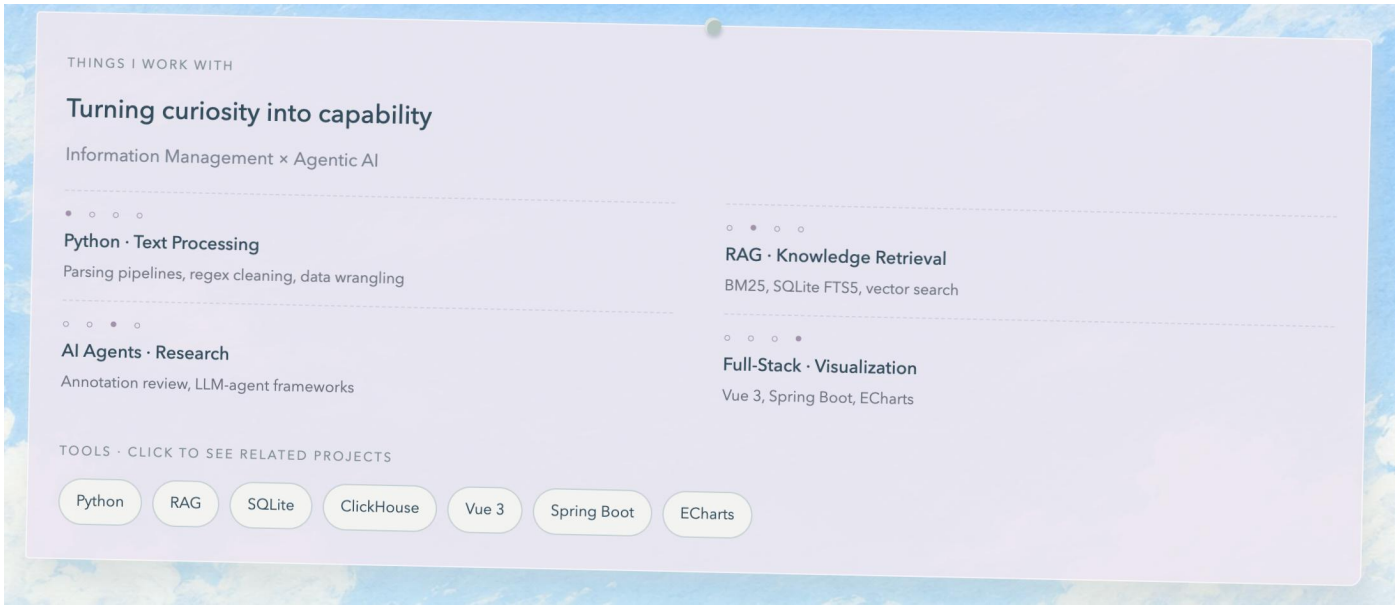
For V0, the Judge rated content and robustness one level above the human assessment but readability one level below, producing a five-point higher total. V1's human and Judge criterion levels agree; the one-point total discrepancy is arithmetic in the raw Judge response. For V2, the Judge rated navigation one level higher, reflecting the later public non-RAG test evidence. The independent Judge is useful as a second reading, not ground truth: it cannot independently click the page, and its judgment depends on the supplied screenshots, source, and test coverage.

3. Task 3: Technique 1 — In-context Few-Shot Learning

V1 was copied from frozen V0 and then changed in its own directory. The prompt supplied examples of component architecture, page skeletons, and gallery organization. It instructed the coding agent to learn structure and interaction patterns only; reference authors' personal text, identity, and images were not copied. The complete prompt revisions, generated source snapshots, and pattern notes are retained in prompt_appendix/task3_v1.

Iteration and examples	Pattern used	Visible V1 effect
Iterations 1–6: al-folio; Shaoyan Liu academic-homepage; Jiya Batra ArtPortfolio-jtc; Michael Kolesidis personal site; Scrapbook Portfolio; Photography Wall; Korbinian Möller template; Hitesh Nandakumar portfolio.	Academic hierarchy, portfolio stories, image-led viewing, accessible controls, restrained motion, and expandable detail.	These references informed scannable project stories, clear education disclosures, keyboard-operable media viewing, and a journal-like visual structure using Aria's own content.
Iteration 7: adopted astro-bento-portfolio, vcard-personal-portfolio, and zine-moskovskaya-excursia; rejected vivek9patel.github.io, LiveTerm, hackclub/scrapbook, and Dopefolio.	Compare page skeletons before selecting one information architecture.	V1 moved to a persistent identity/navigation rail and a chapter-oriented content pane rather than retaining V0's section shell.
Iteration 10: SmugMug layout families, 60 Minute Sites, Media Gallery Layouts, Master Addons masonry comparison, Justified-Gallery, and flickr-justified-gallery.	Compare grid, collage, justified, masonry, and slideshow gallery patterns.	The Life section became an image-led media wall with selectable preview behavior using the existing personal photos and videos.
Iterations 9, 11, and 12 introduced no new samples.	Reuse reviewed patterns rather than expanding the reference set.	Education detail remains expandable and skills stay visible; Iteration 12 focused on interaction clarity and final component behavior.

The final V1 shell includes a journal-cover Home, education journey and skill ledger, indexed project stories, a Life media atlas, and an open-postcard Guestbook. The V0-to-V1 copy contract compared 48 groups with zero text differences and zero structure errors. This supports the claim that content was retained while the component architecture changed.



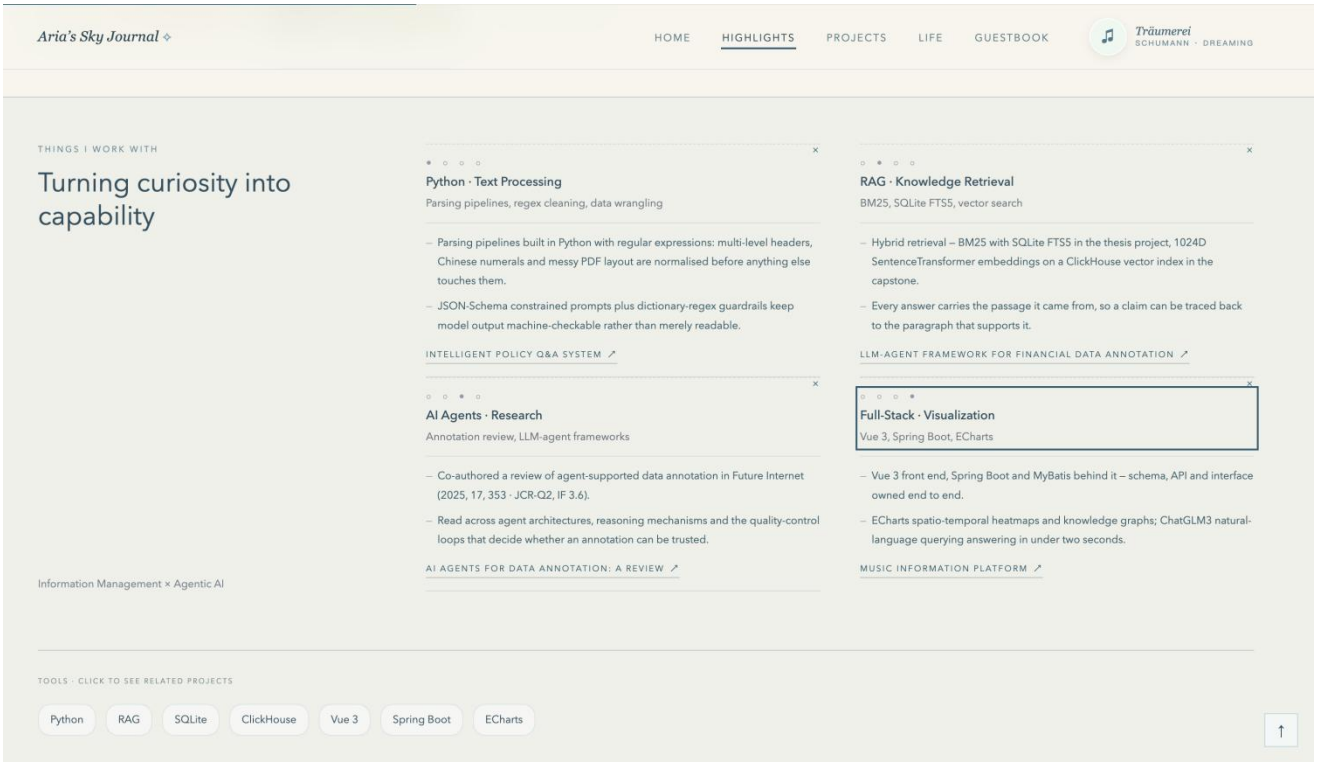


Figure 1. Highlights screenshots supplied for the V0/V1 comparison. Different hierarchy and component treatment are visible; the zero-difference copy contract provides separate evidence of content preservation.

Under the shared desktop E2E protocol, the final V1 R12 run passed 35/35 checks with no page or console errors. It covered navigation, carousel, education and project disclosures, filters, saves, media and video preview, music controls, and empty Guestbook validation. A successful Guestbook submission, length limits, full-site mobile behavior, and a formal contrast audit remain unverified. The independent V1 Judge reported 85/100; its criterion scores recompute to 86/100. The fixed-rubric human assessment is 86/100.

4. Task 3: Technique 2 — Front-end RAG demonstration

V2 is an independent copy of frozen V0. It changes only the course-note question module and does not inherit V1's timeline, skill chips, or page shell. The structured browser-side corpus contains 9,831 records drawn from course notes and project material, with titles, passages, and source metadata. Corpus provenance, query runs, prompts, and repair history are archived under prompt_appendix/task3_v2_evidence.

Workflow and static-site constraint

1. Retrieve. A visitor submits a question. Client-side BM25 returns ranked candidate passages from the local corpus.
2. Filter and rank. All BM25 hits remain inspectable in the closed-by-default reference disclosure. A shared title-led relevance gate and base-concept score prioritize passages whose titles directly match the topic; body-only keyword noise remains evidence but is excluded from answer support.
3. Answer with evidence. The deterministic browser-side composer summarizes only accepted relevant material, attaches citations to the supporting claims, and uses the exact insufficient-materials message when no relevant evidence remains. Reference text is cleaned for visitors; internal prompts and debugging notes are not rendered.

GitHub Pages has no server-side proxy in this project. V2 does not call a live LLM API and contains no API key. Its answer stage is deterministic source-grounded composition, not model generation. BM25 and title rules can miss paraphrases, translation is not always natural, and the implementation cannot provide unrestricted semantic synthesis.

Representative query evidence

Test query	Retrieved evidence and page result	What this demonstrates
How does MapReduce divide and combine work?	7 passages were retrieved. The answer summarized mapping, shuffling, and reducing with source citation [R4]. The folded references retained all hits for inspection.	Answer support is separated from lexical retrieval evidence; the answer cites the topic-bearing passage rather than repeating every BM25 hit.
Tell me about that thing from class.	0 passages; the page displayed "Insufficient reference materials to answer this question."	The unsupported-query path returns a visible answer rather than a blank card or unrelated source.
What is MCP?	7 passages; the answer explains the shared interface and Host/Client roles with citations [R1] and [R3].	Multiple accepted sources can support distinct claims while irrelevant recalled passages stay out of the answer.

A remaining quality issue appears in the Chinese stress queries: agent 是什么、CNN 的开创性意义 currently produce Chinese answers, while the visitor-facing specification calls for English answers. The CNN result is also more compressed than the portfolio-quality answer format intended. The public test confirms retrieval and citations for these cases, but does not establish that the English-answer requirement is met for Chinese input.

5. V2 iterations and effectiveness evidence

The implementation went through documented failures rather than a single rewrite. Early versions exposed prompt text, showed raw or noisy reference content, could display a fallback beside retrieved evidence, and sometimes produced blank or mismatched answers. Later revisions removed prompt/debug material from the visitor UI, put references behind a native disclosure, cleaned displayed passages, unified the relevance and answer branches, filtered title-level noise, corrected citation provenance, and removed query-specific answer catalogs. Dated logs preserve the failures and repairs; they should be read as history, not as descriptions of the final build.

Public test record	Result	Scope and limitation
V2 RAG desktop E2E	9/9 passed	Public Chrome headless at 1512×1000. Covered RAG, MapReduce, memory, vague/out-of-scope, tools, Chinese agent, CNN, and MCP. No RAG console errors.
Non-RAG public E2E	13 functional checks passed	Navigation, education, project detail/filter/save, Life media/photo preview, and Guestbook form. One favicon 404; valid submission/length limits not tested.
Generic RAG regression	28 queries × 10 invariants; all passed	Checks total answers, relevance/fallback, citations, preserved hits, clean UI, answer shape, and provenance. Script-only, not browser E2E.



Figure 3&4. No-evidence states and Current public V2 answer

The latest AI Judge rerun scored V2 91/100 (5, 5, 4, 4), compared with V0's 81/100 (5, 4, 3, 4). The Judge raised Navigation & Task Completion after receiving the fresh public non-RAG evidence. The fixed-rubric human assessment is 86/100 for V2. The same total is not expected because the human score kept navigation at 4 while the Judge assigned 5 based on the supplied checks. A separate user-confirmed five-dimension human scale records V0 79, V1 90, and V2 91; it supplements rather than replaces the official rubric.

6. Findings, trade-offs, and remaining work

V1 shows a visible information-architecture change supported by the few-shot inputs, a zero-difference content contract, and 35/35 desktop checks. The change improves scanning and gives project, education, and media content clearer component roles, at the cost of a more layered and longer page. V2 makes retrieval sources and answer citations inspectable and passes the representative RAG tests. It remains a static demonstration with lexical retrieval and deterministic answers; it cannot be described as a real LLM-based generation service.

The AI Judge totals increased from V0 to both technique versions, but the score change alone does not establish why. The V1 comparison is supported by its component changes and copy/E2E records. The V2 comparison is supported by query-level retrieval/answer evidence, the generic regression record, and public browser checks. Test coverage remains desktop-focused. The V2 non-RAG run's favicon 404 is unresolved, and no broad mobile, accessibility, reload-persistence, or stress audit is claimed.

Submission package and evidence limits

The repository contains independent website/V0, website/V1, and website/V2 copies, the evaluator and raw Judge outputs under evaluation/, and task-labelled prompts, code snapshots, RAG queries, repair logs, and screenshots under prompt_appendix/. This editable DOCX is laid out for A4 PDF export within the eight-page limit. The required combined prompt_appendix.pdf is absent from the checked repository snapshot; the source records remain in folders and still need to be combined.

Available records identify Trae and Codex but do not consistently preserve the underlying model versions; the clean-session Judge records also omit model IDs. None are inferred here. Some Trae history is reconstructed from the user's dated recollection and labelled accordingly in the appendix.

Selected references and provenance

Course materials used as RAG source data include COMP5583 Agentic Computing (Lecture 6, Retrieval-Augmented Generation), COMP5584 LLM for Agent (Lectures 6–7, Tool Use and Agent Memory/RAG), COMP5434 Big Data Computing (Lecture 4, Hadoop and MapReduce), and COMP5205 Introduction to AI (Lecture 5, GNNs and generative models). Exact note filenames, titles, and retrieved excerpts are recorded in the V2 corpus manifest and query archive.

Few-shot component references: al-folio; Shaoyan Liu academic homepage; Jiya Batra ArtPortfolio-jtc; Michael Kolesidis portfolio; Scrapbook Portfolio; Photography Wall; Korbinian Möller template; Hitesh Nandakumar portfolio. Only component structure and interaction patterns were considered; reference identity, wording, and personal assets were not copied.